# MANOJ PAWAR SJ

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## SUMMARY

Backend engineer (SDE 2) with nearly 5 years of experience building distributed, real-time and event-driven systems in Go, Python, TypeScript and Node.js across space tech, e-commerce, gaming and CRM. Currently onsite at ISTRAC, ISRO, building a real-time monitoring and control platform. Cut cloud costs by **40%** with concurrent Go services, made a batch pipeline **6x faster** and removed **90%** of manual ops work with a self-serve scheduler; earlier, lifted e-commerce conversion by **5%** and CRM call pickup rates by **35%**.

## TECHNICAL SKILLS

**Languages:** Go (Golang), Python, TypeScript, JavaScript, SQL
**Backend:** Node.js, Nest.js, Express.js, Hono, REST APIs, Microservices, WebSockets, SSE, MQTT, TCP/IP sockets
**Distributed Systems:** Event-driven architecture, Pub/Sub, Real-time streaming, Rules engines, Binary protocol parsing, RBAC, Caching (Redis)
**Concurrency:** Go goroutines & channels, Python asyncio, threading, multiprocessing
**Databases & ORMs:** PostgreSQL, MySQL, MongoDB, Redis, Firestore, TypeORM, Drizzle, Sequelize, Mongoose
**Cloud & DevOps:** AWS (ECS, EC2, Lambda, SQS, S3, RDS, Route 53), GCP, Docker, Terraform, Gitea CI/CD, Git, Linux (RHEL 9)
**Observability:** Grafana, Loki, OpenObserve (log storage and querying)
**Testing:** pytest, Vitest
**Frontend:** React.js, Next.js, Tailwind CSS
**Other:** SciPy, SGP4, Azure PlayFab, Firebase, Metabase

## PROFESSIONAL EXPERIENCE

**Digantara Industries** | digantara.co.in | Bangalore, India | 10/2023 – Present
**Software Development Engineer 2** | 03/2026 – Present
*Onsite contract engineer at ISTRAC, ISRO | 02/2026 – Present*

- Building backend services in Python and Node.js for a distributed, real-time **monitoring and control platform** across ISRO's ground station network, developed on an **offline, air-gapped RHEL 9 (Linux)** network under ISRO security rules.
- Integrated heterogeneous station hardware over raw **TCP/IP and Telnet sockets**, decoding **binary** and ASCII data streams with bit-level parsing; new device types are added through **configuration alone**, with no code changes.
- Designed a **config-driven rules engine** that evaluates live data for alarm detection, so operators change conditions with **no programming knowledge**.
- Built real-time APIs with **SSE** streaming, user management and **RBAC**, and bandwidth-optimized **MQTT** pub/sub that sends only changed data.

**Software Engineer** | 10/2023 – 02/2026

- Optimized Go APIs that predict satellite positions and check for possible collisions, using **concurrent processing with goroutines and channels** and low memory use, **cutting cloud costs by 40%**.
- Turned business requirements into scalable microservices for 2 commercial products, **FDS** (satellite orbit and manoeuvre planning) and **SPACEMAP** (space object tracking), and set up **Gitea CI/CD** pipelines deploying both to **AWS ECS**.
- Built a self-serve **job scheduler in Nest.js** with role-based access control and user signup, letting teams change job timing without redeploys and **cutting manual ops work by 90%**.
- Rebuilt the batch collision-screening pipeline, cutting each run from **6 hours to 1 hour (6x faster)** through bulk precomputation, early candidate filtering and an optimized search (Python, SGP4, SciPy).
- Built the FDS web app in React.js and Tailwind CSS, with a focus on responsiveness and ease of use.

**Elite Competitions** | elitecompetitions.co.uk | Remote | 02/2023 – 07/2023
**Software Engineer (Full Stack, Contract)**

- Built event-driven communication between services using Nest.js, Next.js, **AWS Lambda and SQS**.
- Built a coupon system (one-time, user-specific and newsletter codes) that raised checkout conversion by **5%**.
- Set up Metabase dashboards tracking business KPIs such as customer lifetime value, cart abandonment rate and average order value.

**Playshifu** | playshifu.com | Bangalore, India | 09/2022 – 01/2023
**Software Engineer (Full Stack)**

- Developed real-time backend services for multiplayer games on **Azure PlayFab**.
- Built internal reporting dashboards in React.js that surfaced **3 major** reporting bottlenecks for the operations team.

**Sunstone** | sunstone.in | Gurgaon, India | 01/2022 – 08/2022
**Software Engineer (Backend)**

- Integrated Airtel and Kaleyra cloud calling into the CRM, raising call pickup rates by **35%**.
- Built the CRM backend's REST APIs in Express.js using MVC architecture.

## PROJECTS

**Gobber** | Go | github.com/manojpawarsj12/gobber
An npm-compatible package installer (package.json, package-lock.json v3) with a global store and symlinked node_modules, concurrent HTTP/2 downloads and integrity checks; benchmarked **~6x faster** than npm on cold installs and **~65x faster** on CI installs.

**smolie-js** | TypeScript | github.com/manojpawarsj12/smolie-js
A reactive UI library under 1 KB with signals, effects and DOM rendering helpers.

**App Tracker** | Python, Plotly, PyPI | github.com/manojpawarsj12/apptracker
A desktop time tracker, published on PyPI, that records time spent per app and website and charts it by day, week or month.

**Vent-Chat** | Node.js, WebSockets, MongoDB
A real-time chat and social app with persistent message history.

## EDUCATION

**B.E. Computer Science and Engineering** | Sathyabama Institute of Science and Technology, Chennai
08/2018 – 07/2022 | CGPA: 8.2/10